ggskewboxplots: Enhanced Boxplots for Skewed Data in R

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Abstract

Traditional boxplots are widely used for summarizing and visualizing the distribution of numerical data, yet they exhibit significant limitations when applied to skewed or heavy-tailed distributions, often leading to misclassification of outliers through *swamping*—flagging typical observations as outliers or *masking*—failing to detect true outliers. This paper addresses these limitations by systematically evaluating several alternative boxplot methods specifically designed to accommodate distributional asymmetry. We introduce `ggskewboxplots`, an R package that integrates multiple robust and skewness-aware boxplot variants, providing a unified and user-friendly framework for exploratory data analysis. Using extensive Monte Carlo simulations under controlled skewness and kurtosis conditions, evaluated across multiple sample sizes and contamination severity levels via the mosaic approach based on the Skewed Exponential Power distribution, we systematically assess the swamping and masking behaviors of each method. Simulation results indicate that classical Tukey-style boxplots are highly prone to swamping and masking, whereas robust skewness-adjusted variants, particularly those leveraging quartile-based skewness measures or medcouple-based adjustments, achieve substantially better performance. These findings offer practical guidance for selecting reliable boxplot methods in applied settings and demonstrate how the `ggskewboxplots` package facilitates accessible, distribution-aware visualizations within the familiar `ggplot2` workflow.

Keywords: outlier, swamping, masking, heavy-tailed

1 Introduction

Boxplots are widely used statistical tools for visualizing the distribution of a variable. First introduced by Spear (1952) and later popularized by Tukey (1977), they provide essential summary statistics such as the median, quartiles, and whiskers, while highlighting potential outliers in a dataset. Due to their simplicity, interpretability, and usefulness in exploratory data analysis, boxplots remain a common tool among researchers and practitioners (Benjamini 1988, Sim et al. 2005). In practice, these plots can be readily produced in both base R (R Core Team 2024), e.g., `boxplot()`, and the `ggplot2` package (Wickham 2016), e.g., `geom_boxplot()`, which together constitute the primary tools for routine boxplot visualization.

However, a fundamental limitation of the traditional Tukey-style boxplot is its reliance on assumptions of symmetry and light-tailedness. It defines outliers as observations falling beyond 1.5 times the interquartile range from the lower and upper quartiles. While this rule performs well under symmetric, mesokurtic distributions like the normal, it tends to over-flag regular observations as outliers — or fail to detect genuine outliers — when the data are skewed or heavy-tailed (Hubert & Vandervieren 2008, Bruffaerts et al. 2014). These issues correspond to common problems in outlier detection known as *swamping* and *masking*. *Swamping* occurs when typical, non-outlying observations are incorrectly identified as outliers, leading to a decrease in specificity or the true negative rate. Conversely, *masking* happens when actual outliers remain undetected because they are obscured by the overall data distribution, reducing sensitivity or the true positive rate. For example, in a right-skewed dataset, the classic boxplot may flag many normal observations on the long tail as outliers, while simultaneously missing subtle but important extreme values. These limitations highlight the need for distribution-aware alternatives that maintain both high sensitivity and specificity across diverse data shapes.

A key question underlying any outlier detection procedure is what characterizes a point as an outlier. Following the distributional characterization adopted in the outlier detection literature (Barnett & Lewis 1994, Rousseeuw & Hubert 2011), we treat an outlier as an observation that deviates from the bulk of the data as generated by an assumed underlying distribution. Under this framework, a point is considered an outlier if it falls in regions of very low probability under the postulated data-generating model. This characterization implicitly assumes that the underlying distribution is unimodal; in multimodal settings, observations in low-density regions between modes may be incorrectly flagged. Throughout this paper, we operate under a unimodal distributional assumption, which is common in practice and consistent with the simulation design described in Section 5.

To address these shortcomings, several alternative boxplots have been proposed. One notable example is the adjusted boxplot by Hubert & Vandervieren (2008), which utilizes the robust medcouple to adjust whisker lengths according to the skewness. Other approaches involve log-transformations (Schwertman et al. 2004), extreme value theory (Bhattacharya et al. 2023), or fully nonparametric variants (Dovoedo & Chakraborti 2015, Walker et al. 2018).

Beyond methodological developments, the growing flexibility of statistical software has also broadened the practical landscape for boxplot customization. In `R`, both base graphics and the `ggplot2` ecosystem allow users to override or fully redefine the statistical components of a boxplot—such as whisker limits, quartile definitions, or outlier rules—through user-supplied summaries or custom statistical layers. For example, the functions `bxp()` and `ggproto`-based extensions make it possible to construct boxplots using arbitrary formulas rather than the classical Tukey definition. However, despite this technical flexibility, the statistical logic of a boxplot cannot be modified directly by end users, and manually supplying the required summaries for each alternative method can be cumbersome—especially for users unfamiliar with the diverse boxplot variants proposed in the literature. As a result,

many robust or distribution-aware boxplot methods remain underutilized in applied work, highlighting the need for accessible tools that implement these approaches in a unified and user-friendly manner.

This paper addresses the lack of an integrated R package that provides multiple alternative boxplot methods designed to handle skewed and heavy-tailed data, aiming to offer a practical and user-friendly tool for data analysts facing distributional challenges. To this end, we developed a new R package implementing a range of robust and flexible boxplot methods. Additionally, we systematically compare the performance of these alternative boxplots across varying levels of skewness and kurtosis (Cavus 2025), guiding users in selecting the most appropriate method regardless of the underlying data distribution. Extensive Monte Carlo simulations are conducted under controlled distributional settings to assess the sensitivity and specificity of each method. For performance evaluation, we employ the mosaic approach (Sánchez-Espigares et al. 2018), based on the Skewed Exponential Power Distribution (Zhu & Zinde-Walsh 2009), which enables flexible control over both skewness and tail weight. We chose this approach because, unlike costly simulation designs that rely on multiple separate distributions, it can represent a wide range of distribution families within a unified framework, thereby enhancing the efficiency and generalizability of our analysis. The main contributions of this paper can be summarized as follows:

1. We introduce `ggskewboxplots`, an `R` package that integrates multiple alternative boxplot methods tailored for skewed and heavy-tailed data, providing a unified and user-friendly framework for practitioners.

2. We conduct extensive Monte Carlo simulations across controlled skewness, kurtosis, multiple sample sizes, and contamination severity levels to systematically evaluate the sensitivity and specificity of each method, providing practical decision guidelines for applied researchers.

3. We demonstrate the advantages of these alternative boxplots in mitigating swamping and masking effects, highlighting their practical relevance for exploratory data analysis and outlier detection in diverse datasets.

The remainder of this paper is structured as follows. Section 2 reviews the alternative boxplot methods included in the study. Section 3 outlines the software architecture of `ggskewboxplots`. Section 4 illustrates package usage through empirical use cases. Section 5 details the Monte Carlo simulation design across varying sample sizes and contamination severity levels. Section 6 presents and discusses the simulation results, and Section 7 concludes the paper.

2 Methodology

The classical Tukey boxplot defines outliers as those observations falling below $Q_1 - k \cdot \text{IQR}$ or above $Q_3 + k \cdot \text{IQR}$, where Q_1, Q_2, and Q_3 denote the first, second or median, and third quartiles, respectively, and $\text{IQR} = Q_3 - Q_1$ is the interquartile range, as illustrated in Figure 1. Here, k is a coefficient and is called the whisker coefficient. It controls the size of the fences and is taken as 1.5 mostly in practice. The values are higher or lower than these fences, labeled as outliers. This rule is widely used in practice to identify the outliers in the data due to its simplicity.

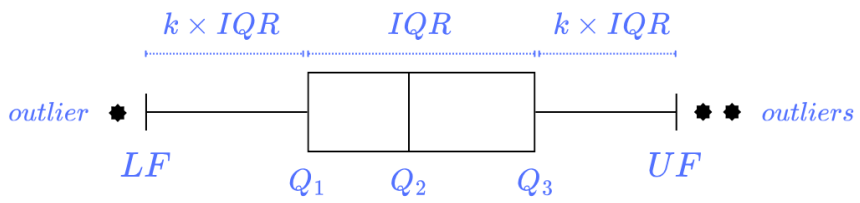


Figure 1: Illustration of a classical Tukey-style boxplot. Observations outside the lower and upper fences $[LF, UF] = [Q_1 - k \cdot \text{IQR},\ Q_3 + k \cdot \text{IQR}]$ are marked as outliers.

However, several modifications have been proposed to accommodate skewed or heavy-tailed data better. These adaptations aim to improve outlier detection performance by adjusting the whisker length or thresholding rules based on distributional characteristics such as skewness and kurtosis. In the following section, we summarize a set of distribution-aware boxplot variations that address these issues.

2.1 Kimber's Boxplot

Kimber (1990) introduced a variation of Tukey's fences that accommodates asymmetry between the lower and upper tails. Instead of relying on a single interquartile range, the approach divides it at the median, resulting in two distinct halves: the lower semi-interquartile range ($SIQR_L = Q_2 - Q_1$) and the upper semi-interquartile range ($SIQR_U = Q_3 - Q_2$). These tailored ranges enable the construction of fences that incorporate the skewness present within the interquartile range.

$$Q_1 - 2k \cdot (Q_2 - Q_1), \quad Q_3 + 2k \cdot (Q_3 - Q_2). \tag{1}$$

This approach maintains the simplicity of the original boxplot while allowing the whisker lengths to reflect asymmetry in the distances between quartiles.

2.2 Hubert's Boxplot

Hubert & Vandervieren (2008) proposed an adjusted boxplot that modifies the whiskers using the *medcouple* (MC), a robust measure of skewness:

$$Q_1 - k \cdot \exp(-3MC) \cdot \text{IQR}, \quad Q_3 + k \cdot \exp(3MC) \cdot \text{IQR} \tag{2}$$

where MC is defined as a robust median difference of right- and left-centered data:

$$MC = \frac{\text{median}(x_i > Q_2) - \text{median}(x_i < Q_2)}{\text{IQR}}. \tag{3}$$

This adjustment results in tighter or wider whiskers depending on the direction and magnitude of the skewness, providing better control over false outlier labeling.

2.3 Adil's Boxplot

Adil et al. (2015) proposed a refinement of the adjusted boxplot by incorporating both the medcouple (MC) and the classical moment-based skewness (SK) into the fence construction. While the earlier method of Hubert & Vandervieren (2008) applied fixed constants as exponents scaled by the medcouple to control the asymmetry of the whiskers, Adil's method replaces these constants with the product of the absolute medcouple and the sample skewness.

The motivation behind this adjustment lies in the behavior of the classical skewness measure: when the skewness is small, the influence on the fence is minimal, resulting in tighter bounds; conversely, larger skewness values expand the fence width, allowing the method to respond to varying degrees of asymmetry in the data adaptively. The modified fences are expressed as:

$$Q_1 - k \cdot \text{IQR} \cdot \exp(SK \cdot |MC|), \quad Q_3 + k \cdot \text{IQR} \cdot \exp(SK \cdot |MC|) \tag{4}$$

where SK is the adjusted skewness statistic, bounded to lie within $[-3.5, 3.5]$ to prevent excessive stretching of the whiskers for extremely skewed distributions. This capping mechanism ensures that the boxplot remains sensitive enough to detect outliers in moderately skewed data without producing overly wide fences that could mask them.

By combining the robustness of the MC with the interpretability of classical skewness, this

method offers a balanced approach to skewness adjustment. In particular, it mitigates the risk of Type II errors in outlier detection, which can occur when overly generous fences fail to flag extreme observations in moderately skewed samples.

2.4 Babura's Boxplot

Babura et al. (2017) used *Bowley's coefficient of skewness* to adjust the whisker lengths:

$$Q_1 - k \cdot \exp(6 \cdot B_C) \cdot \text{IQR}, \quad Q_3 + k \cdot \exp(6 \cdot B_C) \cdot \text{IQR} \tag{5}$$

where Bowley's coefficient is given by

$$B_C = \frac{Q_3 + Q_1 - 2Q_2}{\text{IQR}}. \tag{6}$$

This coefficient captures asymmetry in the quartile positions and is bounded between -1 and 1.

2.5 Walker's Boxplot

Walker et al. (2018) proposed a more flexible variation of the classical Tukey boxplot by introducing a skewness-sensitive formulation. Building on Kimber (1990) idea of using semi-interquartile ranges to define the lower and upper fences asymmetrically, Walker and colleagues incorporated Bowley's coefficient of skewness to enable a more dynamic adjustment.

BC is the quartile-based measure of asymmetry that quantifies the degree and direction of skewness in a distribution. It is defined as the difference between the upper and lower semi-interquartile ranges, normalized by their sum. It ranges from -1 to 1, where values near zero indicate a symmetric distribution, and positive or negative values reflect right

or left skewness, respectively. A greater absolute value of B_C indicates stronger skewness. This coefficient is directly used in the construction of the boxplot fences, using it as a multiplicative adjustment factor. The proposed functional form is as follows:

$$Q_1 - k \cdot \text{IQR} \cdot \frac{1 - B_C}{1 + B_C}, \quad Q_3 + k \cdot \text{IQR} \cdot \frac{1 + B_C}{1 - B_C} \tag{7}$$

This structure allows for more pronounced fence adjustments when B_C deviates substantially from zero. For instance, in the case of right-skewed data where $B_C > 0$, the factor $(1 - B_C)/(1 + B_C)$ becomes less than one. Consequently, the lower fence is extended further outward while the upper fence contracts, leading to a more adaptive outlier detection in skewed distributions.

Walker's method retains the classical Tukey fences for approximately symmetric data, while offering a more realistic delineation of extreme values when skewness is present. This adaptability is especially valuable in domains such as finance or healthcare, where skewed distributions are common and robust outlier detection is critical.

2.6 Junsawang's Boxplot

Junsawang et al. (2021) proposed a dynamic modification of Hubert & Vandervieren (2008) boxplot that incorporates the ratio of semi-interquartile ranges into the exponent:

$$Q_1 - k \cdot \exp\left(B_C \cdot \frac{Q_2 - Q_1}{Q_3 - Q_2}\right) \cdot \text{IQR}, \quad Q_3 + k \cdot \exp\left(B_C \cdot \frac{Q_2 - Q_1}{Q_3 - Q_2}\right) \cdot \text{IQR} \tag{8}$$

This approach flexibly adjusts the whiskers according to local asymmetry around the median, providing robustness across a wide variety of skewed data scenarios.

3 Package Structure

The `ggskewboxplots` package is designed to extend the functionality of `ggplot2` by providing skewness-aware boxplot layers for visualizing asymmetrically distributed data. The central function of the package is `geom_skewboxplot()`, which acts as a drop-in replacement for `geom_boxplot()` but incorporates several alternative methods for adjusting whisker lengths based on data skewness. The statistical transformation logic is handled by a custom `ggproto` object named `StatSkewBoxplot`, which computes five-number summaries and outlier boundaries depending on the selected method, e.g., Tukey, Adil, Hubert, etc. The computation relies on the internal helper function `compute_skew_stats()`, which implements multiple formulae for adjusting whiskers depending on robust measures of asymmetry.

In addition to visualization, the package includes a summarization utility, `summarise_skewbox()`, which returns a tidy table of modified boxplot statistics. This function is useful for exploratory data analysis or generating custom visual summaries without plotting.

The package is implemented entirely in base R and the tidyverse ecosystem, relying on `ggplot2` for graphical rendering, and `dplyr`, `tidyr`, and `rlang` for data manipulation and tidy evaluation. Table 1 summarizes the core user-facing and internal functions, highlighting their roles within the package.

The choice of `ggplot2` as the rendering backend is deliberate. Its `ggproto` architecture allows the statistical transformation logic—fence computation, skewness adjustment, and outlier flagging—to be encapsulated in a custom `Stat` layer, `StatSkewBoxplot`, making `geom_skewboxplot()` a true drop-in replacement for `geom_boxplot()` rather than a standalone plotting function. This design keeps the interface minimal and familiar for the large community of `ggplot2` users, and ensures full compatibility with standard `ggplot2` features

such as faceting, theming, and aesthetic mappings. Furthermore, any plot produced by `geom_skewboxplot()` can be converted to an interactive graphic with a single additional call to `plotly::ggplotly()`, so users who require interactivity are not precluded from obtaining it.

Table 1: An overview of the main and supporting functions in `ggskewboxplots`.

Function	**Purpose**
`geom_skewboxplot()`	Main user-facing function; adds a skew-aware boxplot layer to `ggplot2`.
`compute_skew_stats()`	Internal helper; computes five-number summary and whisker bounds.
`summarise_skewbox()`	Returns modified boxplot statistics in a tidy tibble format.

4 Use Cases

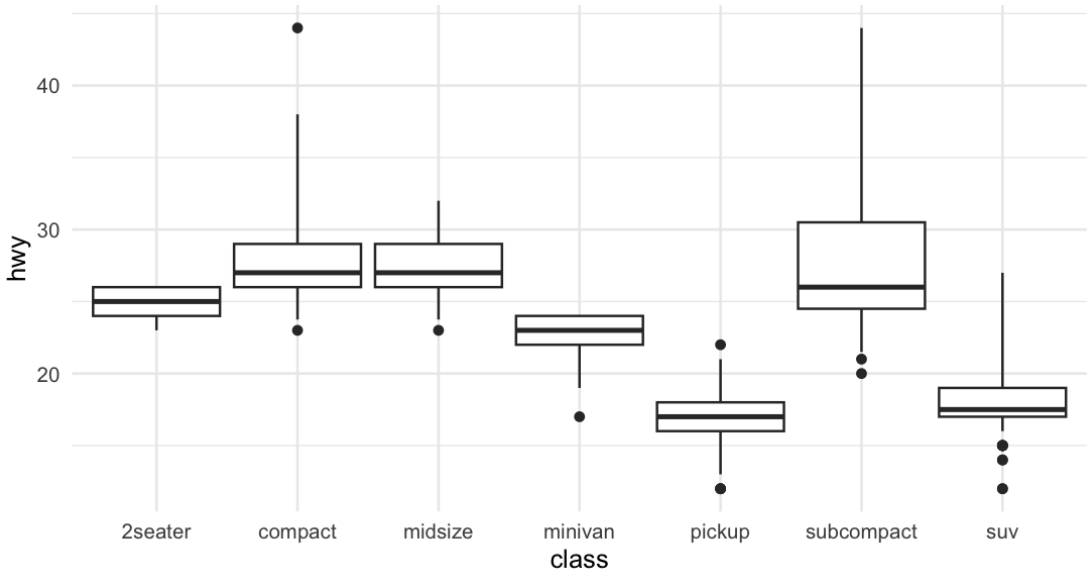
The `ggskewboxplots` package offers a practical and flexible interface for visualizing asymmetric data distributions using skew-aware boxplots. The primary function, `geom_skewboxplot()`, seamlessly integrates with the `ggplot2` ecosystem, allowing users to replace `geom_boxplot()` with minimal adjustments, while benefiting from distribution-aware whisker computations.

4.1 Basic Usage

The simplest use case involves applying one of the available methods, e.g., `adil`, `hubert`, `walker`, directly to a boxplot:

```
> library(ggplot2)
```

```
> library(ggskewboxplots)
> ggplot(mpg, aes(x = class, y = hwy)) +
>  geom_skewboxplot(method = "walker") +
>  theme_minimal()
```

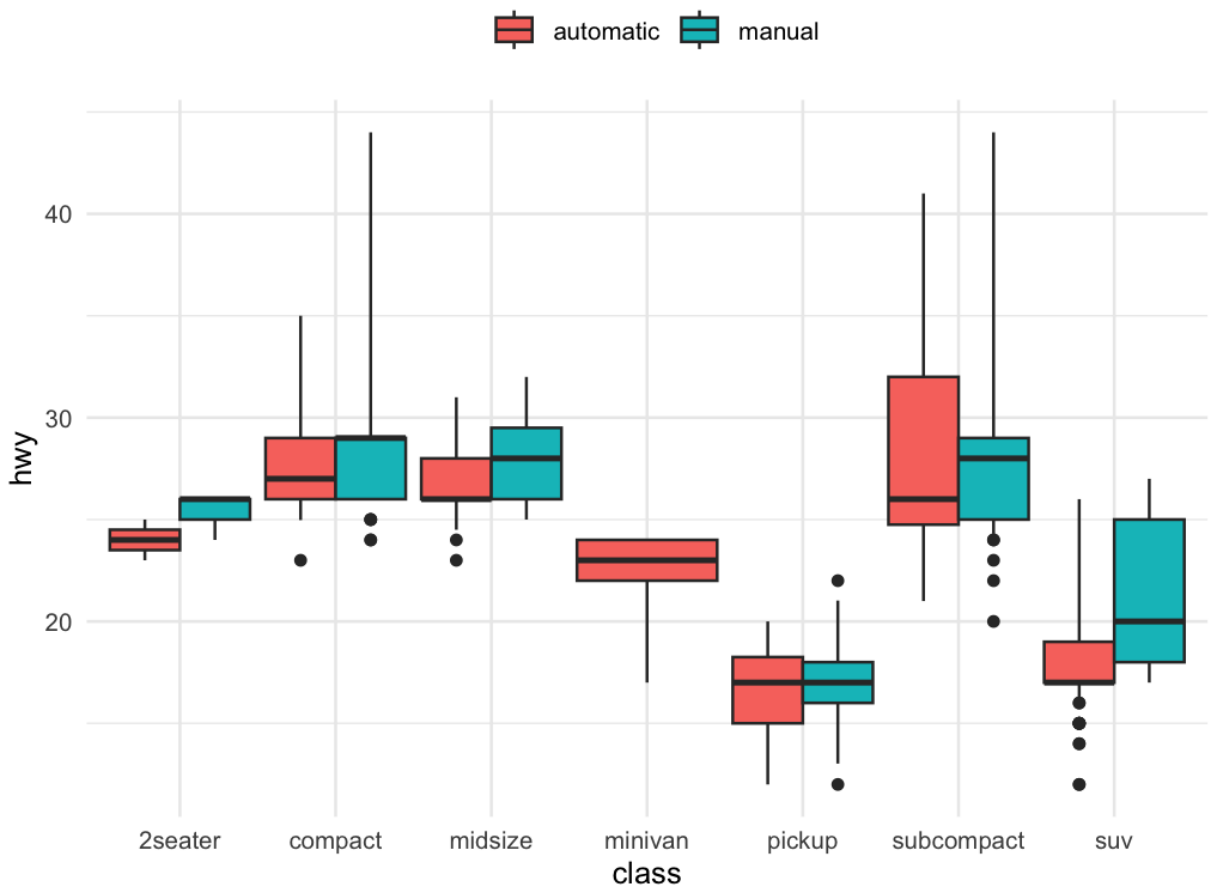


This example replaces the default Tukey boxplot logic with the method proposed by Walker et al. (2018), which adjusts whisker length based on Bowley's coefficient. It allows for improved outlier detection when data are skewed.

4.2 Grouped Comparisons

The function works seamlessly with grouped data, making it suitable for comparing subgroups:

```
> ggplot(mpg, aes(x = class, y = hwy, fill = class)) +
>   geom_skewboxplot(method = "adil") +
>   theme_classic()
```



Here, Adil et al. (2015)'s method adjusts whisker extent using both skewness and the median couple, providing a more robust visualization in the presence of long-tailed distributions.

4.3 Integration with `dplyr` and `summarise()`

The package also includes a summarization function `summarise_skewbox()`, useful for programmatically extracting adjusted boxplot statistics:

```
> library(dplyr)
> mpg |>
+   group_by(class) |>
+   summarise_skewbox(hwy, method = "hubert")
```

```
# A tibble: 7 × 7
  class       ymin lower middle upper  ymax outliers
  <chr>      <dbl> <dbl>  <dbl> <dbl> <dbl> <list>
1 2seater     24.0  24     25    26      26 <int [1]>
2 compact     26.0  26     27    29      44 <int [10]>
3 midsize     25.9  26     27    29      32 <int [6]>
4 minivan     21.9  22     23    24      24 <int [2]>
5 pickup      16.0  16     17    18      22 <int [7]>
6 subcompact  24.5  24.5   26    30.5    44 <int [9]>
7 suv         16.9  17     17.5  19      27 <int [12]>
```

This produces a tidy tibble with modified lower and upper whiskers, central tendency measures, and outlier identification, which can be used for custom visualizations or reporting.

In the resulting table, the `outliers` column stores a list-column containing the specific numerical values or integer indices of anomalous observations flagged as outliers for each subgroup. Practitioners can easily unnest or inspect these flagged values for downstream data cleaning or post-hoc auditing using standard `tidyr` functions:

```
> library(tidyr)
> mpg |>
+   group_by(class) |>
+   summarise_skewbox(hwy, method = "hubert") |>
+   select(class, outliers) |>
+   unnest(outliers) |>
+   head(6)

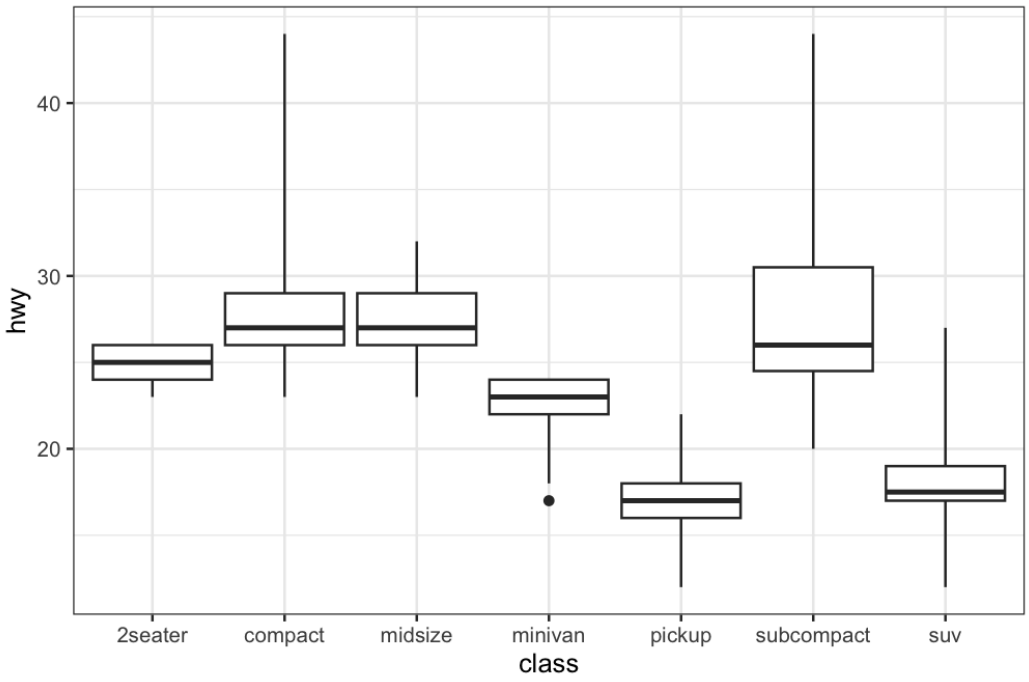
# A tibble: 6 × 2
```

```
  class   outliers
  <chr>      <int>
1 2seater       23
2 compact       25
3 compact       25
4 compact       25
5 compact       25
6 compact       25
```

4.4 Parameter Sensitivity

The tuning parameter `k` can be adjusted to control the whisker extent, allowing further customization depending on the context:

```
> ggplot(mpg, aes(x = class, y = hwy)) +
+   geom_skewboxplot(method = "babura", k = 2) +
+   theme_bw()
```



A larger `k` reduces the number of detected outliers, while a smaller value increases sensitivity. This makes the method adaptable to domain-specific tolerance for extreme values.

5 Simulations

This section presents the design and implementation of simulation studies conducted to evaluate the performance of various alternative boxplot methods under different distributional scenarios. Since traditional boxplots struggle with skewed and heavy-tailed data, we aim to assess how well the adjusted boxplots can control false detection rates of outliers across a wide range of skewness and kurtosis levels.

Simulations are based on synthetic datasets generated from the Skewed Exponential Power Distribution, which offers flexible control over skewness and tail behavior. The mosaic method is employed to visualize the performance across a grid of parameter combinations, providing an intuitive and comprehensive assessment framework.

5.1 Design

To evaluate the performance of various boxplot methods in detecting outliers under different distributional conditions, we designed two complementary simulation scenarios targeting distinct types of classification errors: *swamping* and *masking*. These simulations aim to assess the specificity or true negative rate and sensitivity or true positive rate of each method, respectively.

Let $x_1, \ldots, x_n$ denote an independent sample of size n drawn from the underlying SEPD($\mu = 0, \sigma = 1, \alpha, p$) distribution, and let $[L, U]$ denote the lower and upper fences computed by a given boxplot method on the sample. The two simulation scenarios are formally defined as follows:

5.1.0.1 Swamping Scenario (False Positive Rate):

In the swamping setting, samples contain no artificially injected outliers. Under this setup, any observation falling outside the interval $[L, U]$ constitutes a false positive, defined as a regular observation falsely flagged as an anomaly. The swamping rate for a given replication is defined as:

$$\text{Swamping Rate} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\left(x_i < L \ \vee \ x_i > U\right) \tag{9}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. A high swamping rate signifies that the method's fences are too narrow or insufficiently adaptable to tail behavior, resulting in an unacceptably high type I error rate.

5.1.0.2 Masking Scenario (Contamination Mechanism and False Negative Rate):

In the masking setting, synthetic outliers are injected into the sample. Following the contamination scheme of Junsawang et al. (2021), the uncontaminated sample observations are first sorted in ascending order to obtain the order statistics $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$. Contamination is then applied to both tails with a contamination fraction of $\varepsilon = 0.05$. Specifically, $k = \max(1, \lfloor n\varepsilon/2 \rfloor)$ observations in each tail are multiplied by a scaling factor $c \in \{2, 4, 6\}$:

$$\begin{aligned} x^*_{(i)} &= c \cdot x_{(i)}, \quad i = 1, \ldots, k \quad \text{(lower-tail outliers)}, \\ x^*_{(n-j+1)} &= c \cdot x_{(n-j+1)}, \quad j = 1, \ldots, k \quad \text{(upper-tail outliers)}. \end{aligned} \tag{10}$$

The set of true synthetic anomalies is thus $\mathcal{O} = \{x^*_{(1)}, \ldots, x^*_{(k)}, x^*_{(n-k+1)}, \ldots, x^*_{(n)}\}$ with cardinality $|\mathcal{O}| = 2k$. The scaling factor c controls the extremity of the contamination: $c = 2$ represents mild contamination where anomalies remain close to the data boundary, $c = 4$ corresponds to moderate contamination, and $c = 6$ reflects severe contamination.

After recomputing the fences $[L, U]$ on the contaminated sample, the masking error, defined as the proportion of genuine outliers that remain undetected within the whiskers, is given by:

$$\text{Masking Error} = 1 - \frac{1}{2k} \sum_{x \in \mathcal{O}} \mathbb{I}\left(x < L \ \vee \ x > U\right). \tag{11}$$

A masking error of 0 indicates perfect outlier detection with 100% sensitivity, whereas higher values indicate that true anomalies have been masked by excessively wide whiskers.

We performed simulations using the mosaic method, detailed in Section 5.2, on a 49×49 parameter grid covering a broad spectrum of skewness (α) and kurtosis (p) values. Simulations were carried out across sample sizes $n \in \{20, 50, 100\}$, conducting 10,000 Monte Carlo replications for every grid cell.

In the resulting mosaic plots, each tile displays the Monte Carlo average of the respective classification error, reporting the swamping rate in Figure 3 and the masking error in Figures 4–5 and Appendix Figures 6–7. Tiles are color-coded using the Viridis palette, ranging from dark purple representing optimal performance at 0% to bright yellow reflecting elevated error at 10% or higher. Consequently, darker regions consistently indicate superior performance across both scenarios.

5.2 Mosaic Method

The mosaic method (Sánchez-Espigares et al. 2018) is a visual framework for summarizing the performance of statistical procedures across a wide range of distributional conditions. It works by partitioning a plot into $k \times k$ subgraphs, where each subgraph corresponds to a unique combination of parameters governing a distribution family. In our study, each subgraph represents a simulation result under a specific combination of skewness and kurtosis levels based on the Skewed Exponential Power Distribution (SEPD) (Zhu & Zinde-Walsh 2009). The probability density function of SEPD is defined as:

$$f(x \mid \mu, \sigma, \alpha, p) = \begin{cases} \frac{1}{K(p)} \exp\left(-\frac{1}{p}\frac{|x-\mu|^p}{2\alpha\sigma^p}\right), & x \leq \mu \\ \frac{1}{K(p)} \exp\left(-\frac{1}{p}\frac{|x-\mu|^p}{2(1-\alpha)\sigma^p}\right), & x > \mu \end{cases} \tag{12}$$

where $K(p) = \frac{1}{2}p^{1/p}\Gamma\left(1+\frac{1}{p}\right)$ is a normalizing constant, μ is the location parameter, σ the scale, $\alpha \in [0,1]$ controls skewness, and $p > 0$ controls tail weight. When $\alpha = 0.5$, the distribution is symmetric; as $p \to \infty$, the distribution becomes more uniform; smaller p values result in heavier tails.

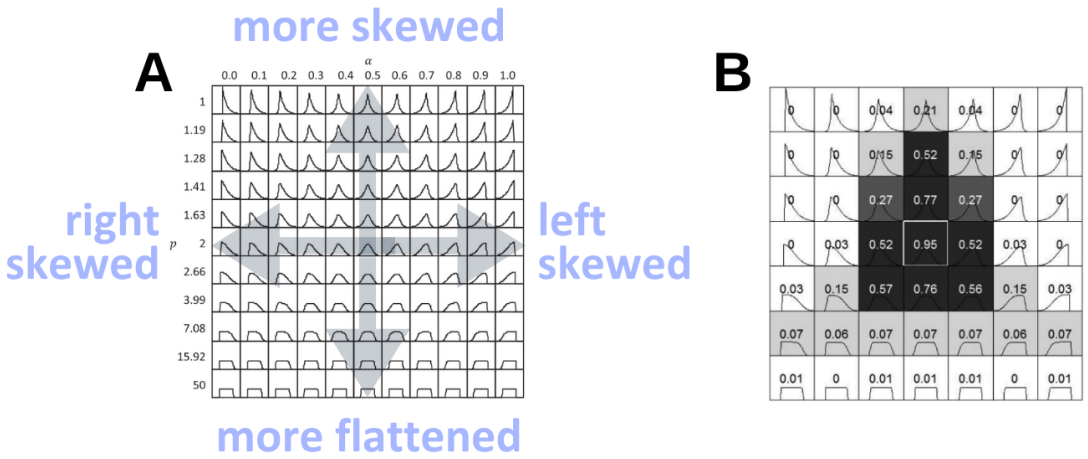


Figure 2: Illustration of the mosaic approach. **Panel A**: Shape changes in SEPD as α (horizontal axis) and p (vertical axis) vary, visualizing transitions from right-skewed to left-skewed distributions horizontally and decreasing kurtosis vertically. **Panel B**: Simulated performance of an outlier detection method across the same parameter grid. Each tile shows the proportion of false positives or true positives for a specific (α, p) pair, where darker tiles indicate lower error rates and lighter tiles indicate poorer performance.

This approach is illustrated in Figure 2, which allows us to evaluate and compare the robustness of boxplot variants across a broad and continuous range of distributional shapes, rather than relying on a limited set of canonical distributions such as the normal and

exponential. It also provides a cost-effective alternative to enumerating dozens of separate simulations for each distributional family, as the SEPD can flexibly approximate many common unimodal, skewed, and heavy or light-tailed distributions.

6 Results

This section presents a comparative evaluation of several boxplots under two challenging scenarios: swamping and masking. Swamping refers to the incorrect flagging of regular observations as outliers, whereas masking denotes the failure to identify true outliers hidden within the bulk of the data. For both scenarios, extensive simulations were conducted across a 49×49 grid of distributional settings, and the outcomes are visualized using mosaic plots. These results provide a systematic assessment of how each method responds to varying degrees of skewness, tail weight, and sample size.

6.1 Swamping

Figure 3 compares the resistance of the seven boxplot methods against swamping across sample sizes $n = 20, 50$, and 100. Tiles are color-coded using the Viridis scale, where dark purple represents a false positive rate near 0%, indicating optimal swamping resistance, and bright yellow corresponds to false positive rates reaching or exceeding 10%, reflecting an elevated risk of falsely classifying regular observations as outliers.

Tukey (1977)'s classical boxplot displays predominantly dark purple tones across large portions of the parameter grid for all sample sizes, maintaining an overall false positive rate below 2.6%. However, lighter greenish-yellow patches emerge along the lateral margins representing extreme skewness and in the lower rows representing platykurtic distributions, indicating that standard fences become somewhat porous when the data deviate substantially from normality. As the sample size increases to $n = 50$ and $n = 100$, the dark central zone

expands, reflecting a more stable fence estimation around moderate skewness.

Kimber (1990)'s method shows a swamping profile very similar to Tukey's, with low false positive rates in symmetric and moderately skewed configurations, achieving a mean swamping rate of 2.9% at $n = 20$ that drops to 1.5% at $n = 100$. The use of asymmetric semi-interquartile fences effectively protects the bulk of the data, although some swamping persists at the extreme corners representing simultaneously flat and heavily skewed distributions.

Hubert & Vandervieren (2008)'s adjusted boxplot leverages the medcouple statistic and produces very low swamping rates across the central domain of the mosaics. Because the medcouple scaling factors dynamically widen the fences in the direction of skewness, Hubert's method effectively suppresses false detections in moderately skewed regimes. At $n = 20$, slight yellowing occurs in platykurtic and extreme skewness boundaries, but as sample size grows to $n = 100$, the entire mosaic shifts almost uniformly toward deep purple tones, with a mean swamping rate of 2.0%.

Adil et al. (2015)'s modification combines the medcouple with classical skewness and achieves one of the lowest false positive rates among the adjusted methods, recording a mean swamping of 2.6% at $n = 20$ and 2.0% at $n = 100$. Dark purple regions dominate across both symmetric and moderately skewed distributions, with a particularly pronounced resistance to swamping in leptokurtic settings.

In contrast, Babura et al. (2017)'s method exhibits a notably higher susceptibility to swamping, displaying widespread bright yellow regions across the mosaics, with a mean swamping rate of 10.8% at $n = 20$ and 5.9% at $n = 100$. Because Babura's formulation applies substantial asymmetric corrections based on Bowley's coefficient, it tends to contract the opposing fence excessively, leading to an elevated rate of falsely flagged non-outliers in flat and highly skewed distributions.

Walker et al. (2018)'s method, while also employing Bowley's coefficient of skewness, incorporates a more conservative ratio-based adjustment that avoids the over-contraction observed in Babura. Consequently, Walker's mosaics remain overwhelmingly dark purple across moderate skewness and platykurtic conditions, achieving excellent swamping control with mean swamping of 3.7% at $n = 20$, dropping to 1.4% at $n = 100$.

Junsawang et al. (2021)'s method dynamically recalibrates Hubert's fences using semi-interquartile ratios and delivers consistently low false positive rates across the parameter spectrum, averaging 3.9% swamping at $n = 20$ and 2.2% at $n = 100$. It successfully avoids unwarranted false alarms across most unimodal and skewed configurations, with only modest swamping observed in the extreme platykurtic corners.

Overall, these swamping results confirm that while conservative methods such as Tukey, Kimber, Adil, and Walker tightly constrain false detections across the vast majority of regular distributions, methods with overly aggressive fence contractions such as Babura incur a discernible swamping penalty, particularly in small samples.

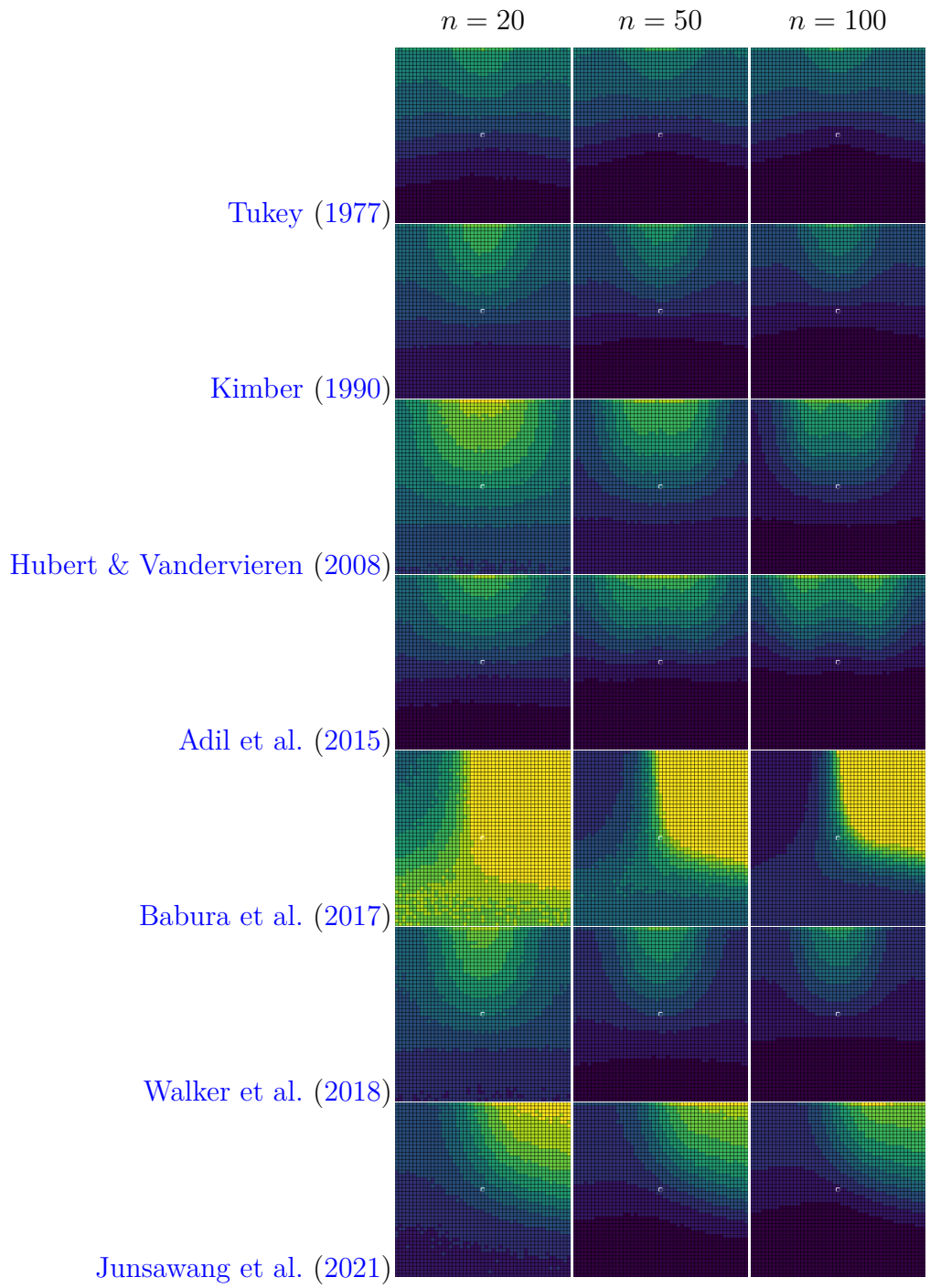


Figure 3: Simulation results for the *swamping* scenario across the 49×49 parameter grid for sample sizes $n \in \{20, 50, 100\}$. Tiles indicate false positive rates color-coded via the Viridis palette from 0% (dark purple, optimal resistance) to 10% or higher (bright yellow, elevated swamping).

6.2 Masking

Figure 4 compares the performance of the seven outlier detection methods across sample sizes $n = 20, 50$, and 100 under the mild masking scenario where $c = 2$. In these mosaic plots, tiles are color-coded using the Viridis scale to represent masking severity, defined as the proportion of undetected genuine outliers computed as $1 -$ detection rate. Dark purple tiles indicate near-zero masking error, representing excellent sensitivity where virtually all outliers are detected, whereas bright yellow tiles correspond to masking rates reaching or exceeding 10%, reflecting scenarios where subtle anomalies remain concealed within the data bulk.

Under mild contamination where $c = 2$, synthetic outliers are generated close to the boundary of the regular observations, creating an inherently challenging detection task. Consequently, at the smallest sample size, $n = 20$, all seven methods exhibit widespread bright yellow mosaics, with mean masking error rates ranging from 25.0% to 42.8%. Because outlier signals are subtle and sample variability is high, standard and adjusted fences alike struggle to reliably isolate genuine anomalies at $n = 20$.

As the sample size increases to $n = 50$ and $n = 100$, substantial differences emerge in how effectively the methods overcome mild masking. Kimber (1990)'s and Tukey (1977)'s methods exhibit the fastest emergence of dark purple zones as the sample size grows. Kimber achieves the lowest average masking severity at $n = 50$, recording 15.9%, and at $n = 100$, recording 13.8%, followed closely by Tukey with 18.8% at $n = 50$ and 17.7% at $n = 100$. Because these methods maintain tighter, more rigid fences, they are quick to flag observations exceeding the empirical quartiles once sample estimates stabilize. However, as noted in Section 6.1, this higher sensitivity comes at the expense of elevated swamping in skewed settings.

Walker et al. (2018)'s method similarly demonstrates a rapid transition toward darker purple

and blue tones in the central and moderately skewed regions, reducing average masking error to 15.8% at $n = 100$. Walker's ratio-based adaptation achieves an advantageous balance, delivering robust detection without inducing excessive false positives. Meanwhile, Hubert & Vandervieren (2008)'s and Junsawang et al. (2021)'s methods achieve intermediate masking resistance, displaying expanding dark bands across symmetric and moderately skewed regimes, with mean masking errors of 17.1% and 19.4% at $n = 100$, respectively. In both approaches, the medcouple-driven fence expansion slightly reduces sensitivity to mild outliers in exchange for superior resistance to swamping under asymmetric distributions.

In contrast, Babura et al. (2017)'s and Adil et al. (2015)'s methods exhibit persistent bright yellow and light green tiles even at $n = 100$, with mean masking rates remaining at 35.0% and 38.3%, respectively. Because both methods implement aggressive adjustments that substantially widen the outer fences in skewed directions, mild outliers with $c = 2$ frequently remain enclosed within the whiskers and go undetected.

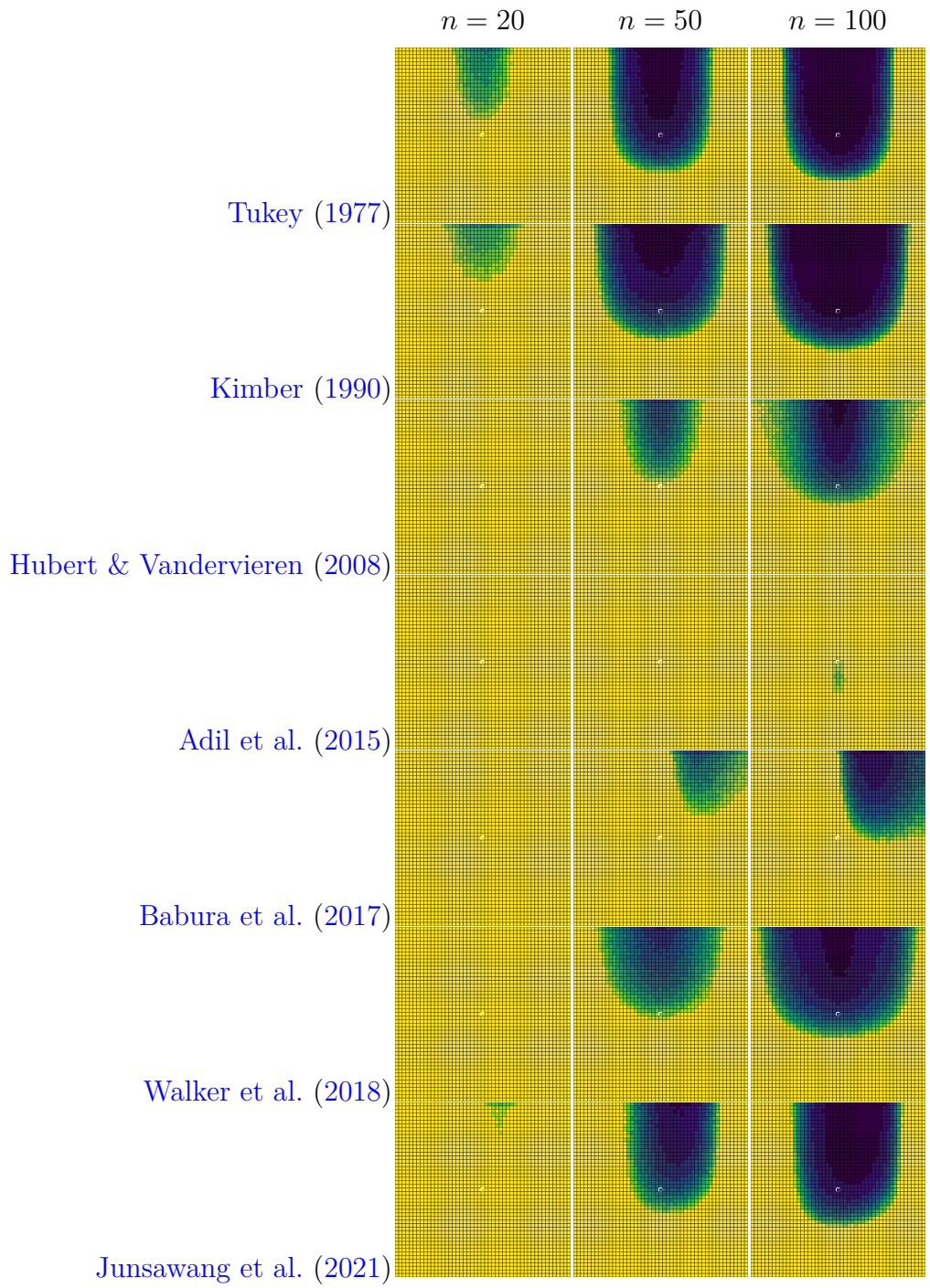


Figure 4: Simulation results for the *masking* scenario under mild contamination ($c = 2$) across sample sizes $n \in \{20, 50, 100\}$. Darker tiles reflect lower masking errors (higher outlier detection sensitivity).

6.3 Impact of Contamination Severity

To further investigate how outlier extremity affects masking resistance, Figure 5 displays the performance of all seven methods at sample size $n = 20$ across three contamination severity levels: mild ($c = 2$), moderate ($c = 4$), and severe ($c = 6$). Complementary results for larger sample sizes ($n = 50$ and $n = 100$) are reported in Appendix Figures 6 and 7.

As the scaling factor increases from $c = 2$ to $c = 6$, the synthetic outliers are shifted farther into the distribution tails, making them progressively easier to distinguish from uncontaminated observations. Consequently, all methods exhibit darker mosaic tiles under larger c values. However, pronounced differences emerge across methods in how rapidly sensitivity improves.

Under mild contamination where $c = 2$, anomalous observations lie close to the normal data mass, causing all methods to experience substantial masking, as indicated by the predominance of bright yellow tiles across the first column of Figure 5. When the contamination severity increases to moderate ($c = 4$), the outliers are shifted farther into the tails. Standard Tukey and Kimber boxplots immediately transition to solid dark purple mosaics with less than 0.1% masking error, because their fixed whiskers are rapidly surpassed by the enlarged anomalies. Methods with moderate adaptive adjustments, such as Walker et al. (2018) with 5.1% mean masking, Junsawang et al. (2021) with 3.8%, and Hubert & Vandervieren (2008) with 7.4%, also become predominantly dark purple, successfully detecting the vast majority of outliers while preserving skewness adaptation. In contrast, methods with heavy asymmetric expansions (Adil et al. (2015) and Babura et al. (2017)) still retain visible yellow regions, especially in asymmetric and heavy-tailed distributions.

Under severe contamination where $c = 6$, the outliers are placed far into the extreme tails. Almost all robust methods, including Hubert, Walker, and Junsawang, achieve near-zero masking error (1.2% to 2.2%) across virtually the entire parameter space. Even

Adil et al. (2015) and Babura et al. (2017) experience substantial darkening, although their expansive whiskers continue to mask a small subset of extreme anomalies in highly platykurtic configurations.

The same progressive improvement with increasing c is observed at $n = 50$ and $n = 100$, as shown in Appendix Figures 6 and 7. Larger sample sizes further accentuate the performance of skewness-adjusted methods: as empirical quantiles and skewness estimators stabilize, Walker, Junsawang, and Hubert reach near-perfect outlier detection, displaying solid dark purple mosaics under $c = 4$ and $c = 6$, combining rigorous masking resistance with robust swamping protection.

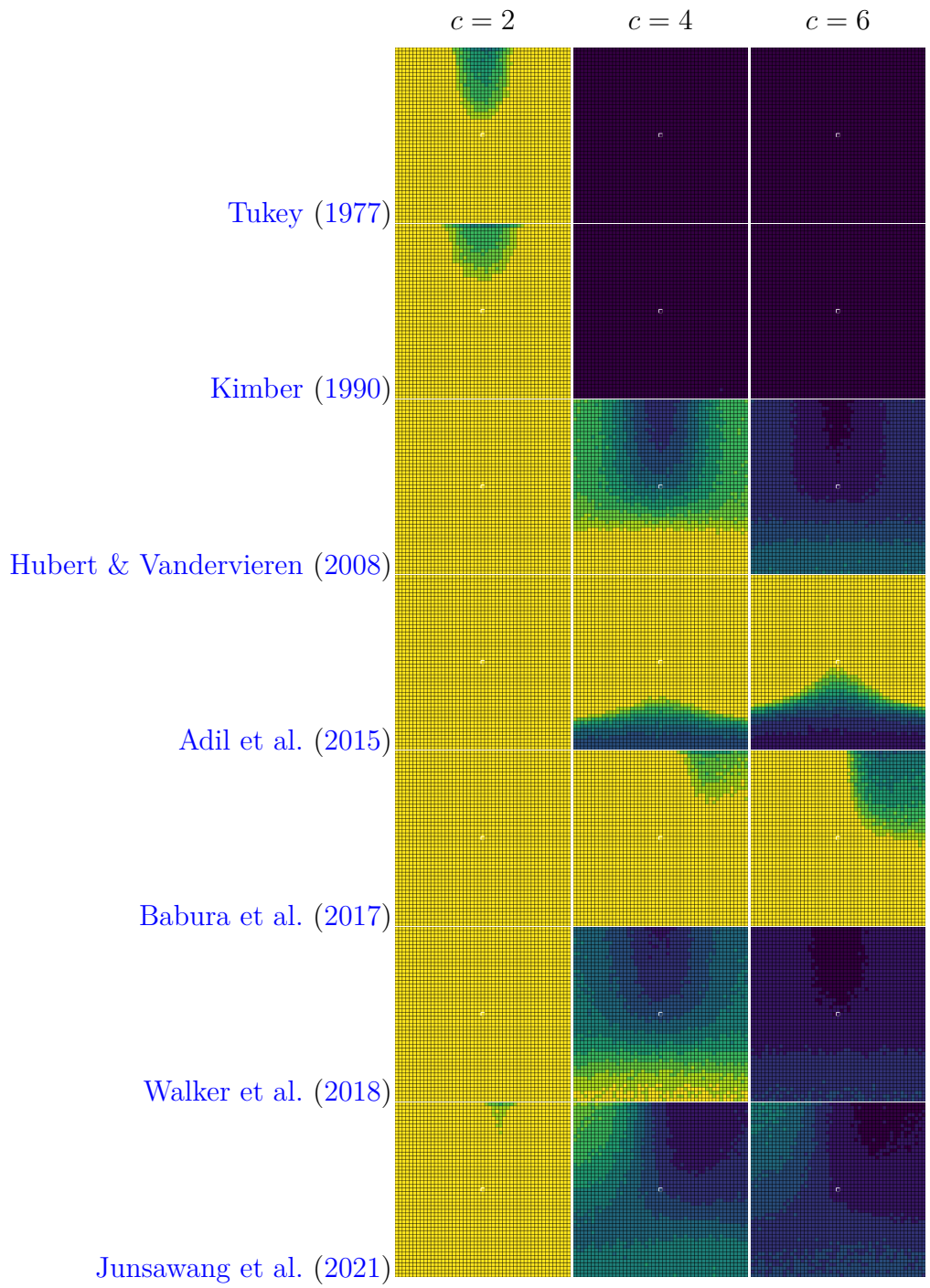


Figure 5: Masking error rates across varying contamination severity levels ($c \in \{2, 4, 6\}$) at sample size $n = 20$. Columns represent contamination levels; rows correspond to boxplot methods.

7 Conclusions

This paper introduced `ggskewboxplots`, an `R` package that integrates a comprehensive collection of skewness-aware boxplot methods for visualizing asymmetric and heavy-tailed distributions. By embedding multiple robust whisker-adjustment strategies into the familiar grammar of graphics (Wickham 2010), the package provides a practical and unified framework for improving outlier detection in exploratory data analysis.

Extensive Monte Carlo simulations conducted under controlled skewness–kurtosis conditions revealed clear performance differences among the examined boxplot variants. The traditional Tukey-style boxplot consistently exhibited high rates of swamping and masking, especially in highly skewed or platykurtic settings, demonstrating its limited reliability for non-symmetric data. Methods incorporating quartile-based skewness measures, such as those proposed by Babura et al. (2017) and Walker et al. (2018), showed the strongest and most consistent performance across both swamping and masking scenarios, retaining robustness even at small sample sizes. Similarly, the dynamic extension by Junsawang et al. (2021) performed well in complex distributional structures by adapting whisker lengths to local asymmetry around the median. Adjusted approaches relying on the medcouple—including those by Hubert & Vandervieren (2008) and Adil et al. (2015)—also improved performance relative to classical methods, particularly in moderately skewed distributions.

Overall, the simulation results underscore that no single boxplot method is uniformly optimal, but robust skewness-aware variants generally provide substantial gains in sensitivity and specificity under asymmetric distributions. The `ggskewboxplots` package translates these methodological improvements into an accessible visualization toolkit, allowing practitioners to select appropriate boxplot forms for their data without leaving the `ggplot2` workflow. Future work may extend the package to multivariate analogues or integrate automatic diagnostic tools that recommend suitable boxplot methods based on empirical skewness

and kurtosis. Together, these contributions enhance the reliability of visual exploratory analysis in real-world settings where deviation from symmetry is the norm rather than the exception.

Disclosure statement

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Supplementary Materials

The `ggskewboxplots` package is available on CRAN: https://cran.r-project.org/package=ggskewboxplots, and the use cases are taken from the vignettes of the package. The complete replication materials, simulation scripts, and precomputed score datasets are publicly available at https://github.com/mcavs/ggskewboxplots-paper.

Appendix: Additional Simulation Results for Contamination Severity

This appendix provides the complete simulation mosaics evaluating masking severity across

mild ($c = 2$), moderate ($c = 4$), and severe ($c = 6$) contamination levels for sample sizes $n = 50$, shown in Figure 6, and $n = 100$, shown in Figure 7.

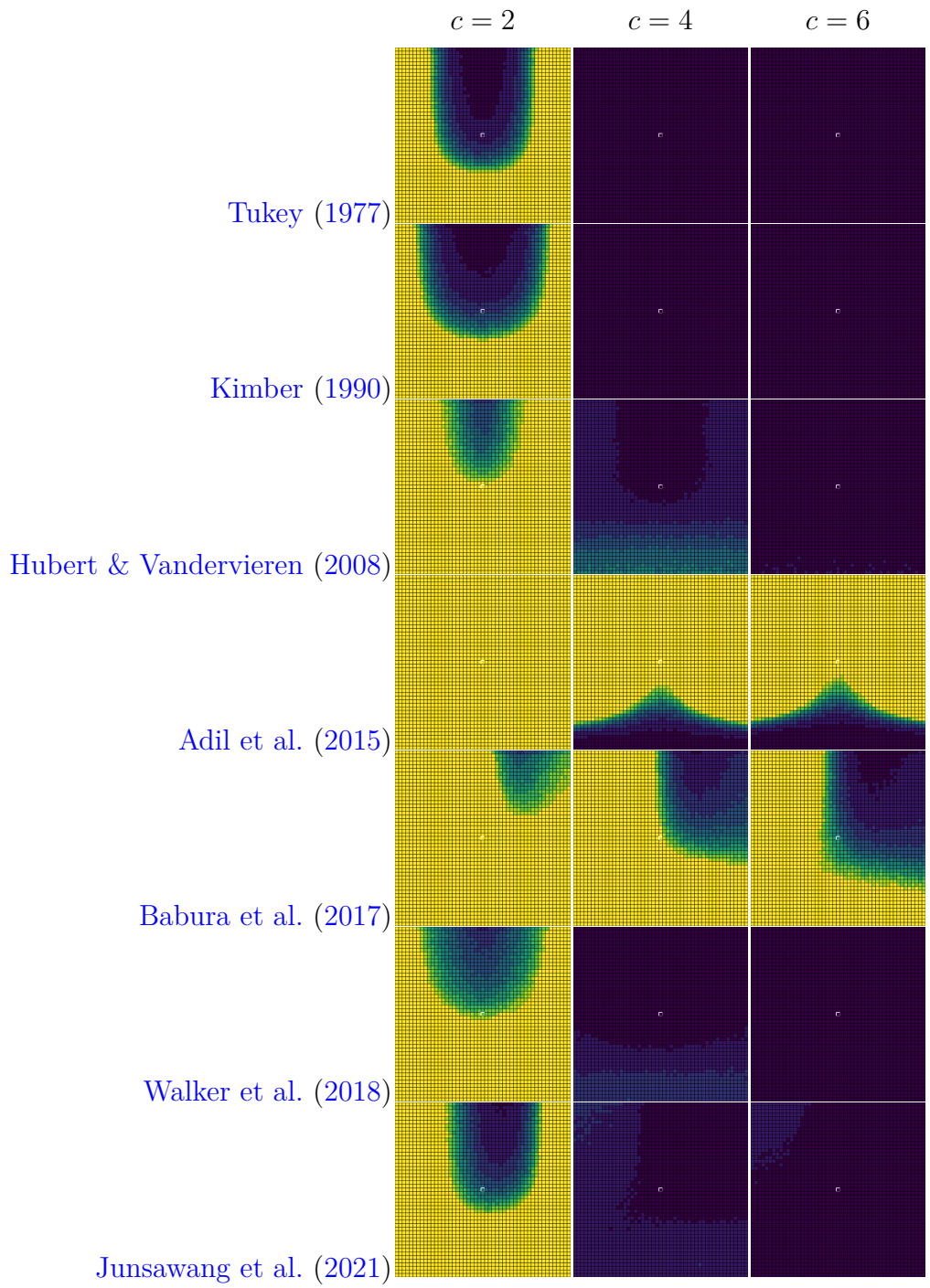


Figure 6: Masking error rates across varying contamination severity levels ($c \in \{2, 4, 6\}$) at sample size $n = 50$. Columns represent contamination levels; rows correspond to boxplot methods.

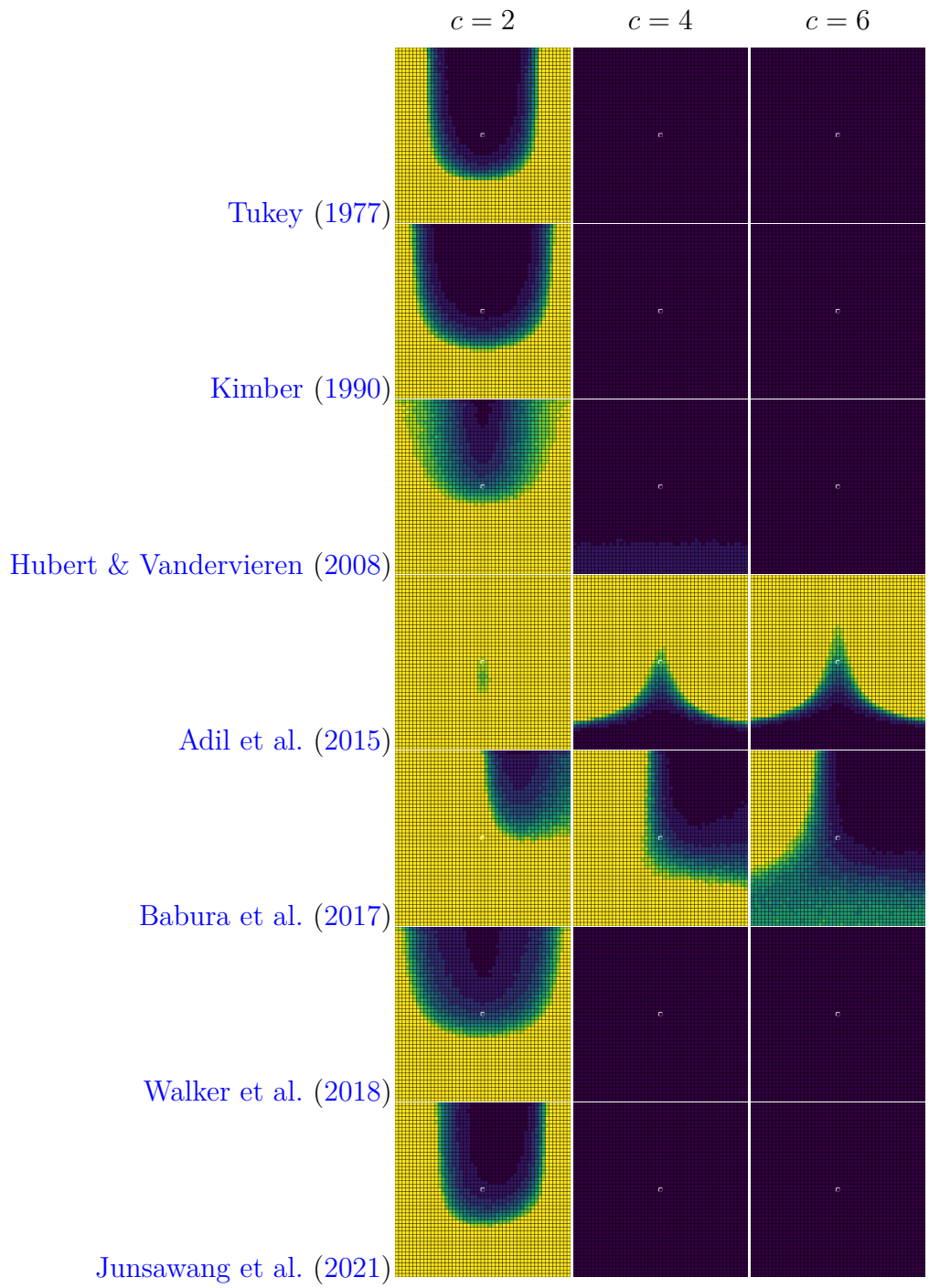


Figure 7: Masking error rates across varying contamination severity levels ($c \in \{2, 4, 6\}$) at sample size $n = 100$. Columns represent contamination levels; rows correspond to boxplot methods.

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